How Dog Traits and Region Affect Time to Adoption in Shelters

Public animal shelter records from Austin, Norfolk, and Sonoma County

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Executive Summary

Goal of the study

- Which dog traits correlate with faster adoption
  - age, region, breed, size, color, sex
- Can models trained on public shelter records accurately predict a dog’s length of stay based on these traits?

Significance

- Shelter staff can optimize limited resources
  - Can allocate extra marketing or foster resources to “low-appeal” or “at risk” dogs
- Will help increase adoption rates and can minimize negative outcomes such as euthanasia

Executive Summary

Data

- Open-data portals: Austin Animal Center (TX), Norfolk Animal Care Center (VA), Sonoma County Animal Services (CA)
- 106,786 raw rows
- Shared outcome window
    - May 5, 2025 to April 5, 2026
- Final sample: 2,884 adopted dogs with known sex and spay/neuter status
- Stay length = outcome date minus intake date, recomputed for every source

Executive Summary

Key Findings

- Puppies adopted fastest everywhere
- ANOVA ranks breed family first (Partial eta^2 = 0.088; p = 6.68e-54)
- Color is not significant (p = 0.281)
- Exact days: unpredictable (Mean Absolute Error near 50)
- Adoption within intervals: predictable and accurate

Data Collection, Cleaning, and Analysis

1. Data collection
Austin, Norfolk, Sonoma gov-data portals; 106,786 animals recorded

2. Cleaning and harmonization
Dogs only, valid dates; grouped breeds, colors, outcomes; 2,884 adopted dogs kept

3. Exploratory analysis
Right-skewed stays; age patterns differ by region

4. Regression and ANOVA
ANOVA to find significant variables for adoption time

5. Prediction models
Use neural network and random forest to predict

6. Conclusions
Age, region, and breed are main factors in influencing outcome

Data cleaning

Raw sources (all animal types)

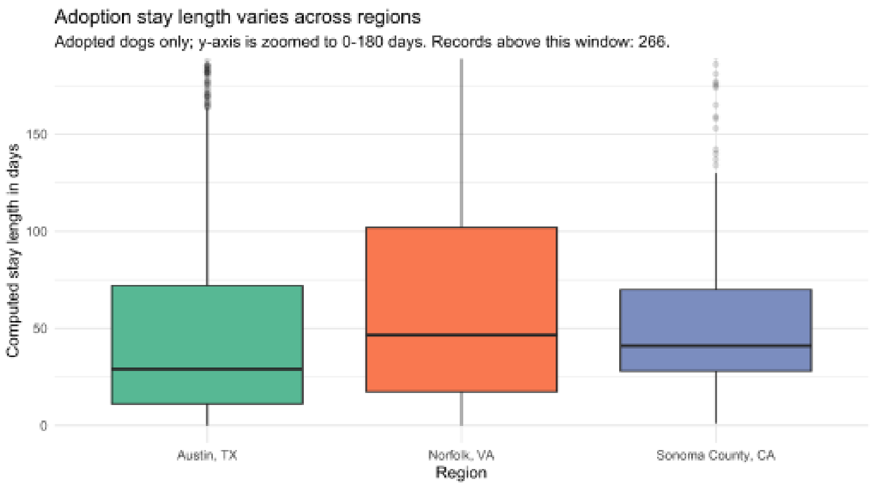
Region	Raw rows	Age information
Austin, TX	13,571	Date of birth
Norfolk, VA	59,540	Years old and months old
Sonoma County, CA	33,675	Approximate date of birth

- One outcome window for all regions: May 5, 2025 to April 5, 2026
- Age from date of birth (Austin, Sonoma) or years + months (Norfolk)
- Ages below 0 or above 25 years dropped as data errors

Data Cleaning

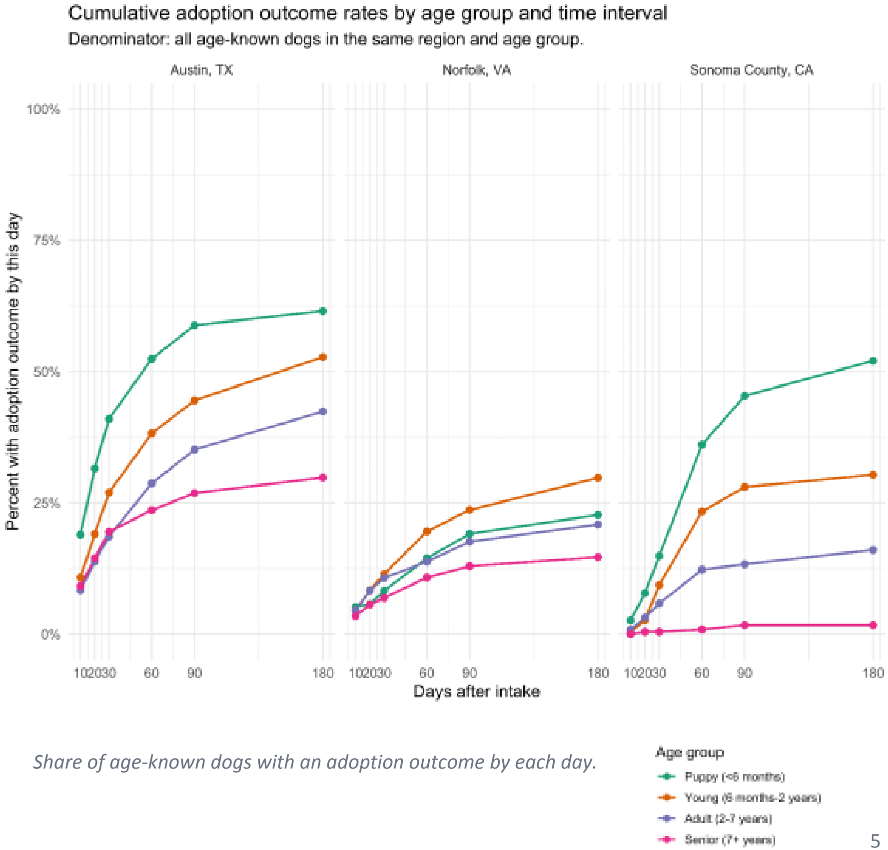
- Removed only to include
  - Dogs only; valid intake/outcome dates; stay > 0 days
- Breeds grouped into 7 families:
  - Hound/Beagle, Retriever, Shepherd, Pitbull/Staffy, Small Companion, Working/Large, Other
- Colors grouped into 8:
  - Black, White, Brown, Tan/Cream, Gray/Blue, Red/Orange, Patterned, Other
- Outcomes grouped into 6:
  - Adoption, Died, Euthanasia, Return to Owner, Transfer, Other
- Adoption outcomes only; return to owner, transfer, euthanasia, death excluded
- Unknown sex or spay/neuter status dropped
- Final sample: 2,884 dogs (2,071 Austin, 486 Norfolk, 327 Sonoma)

EDA: stay length by region & age groups



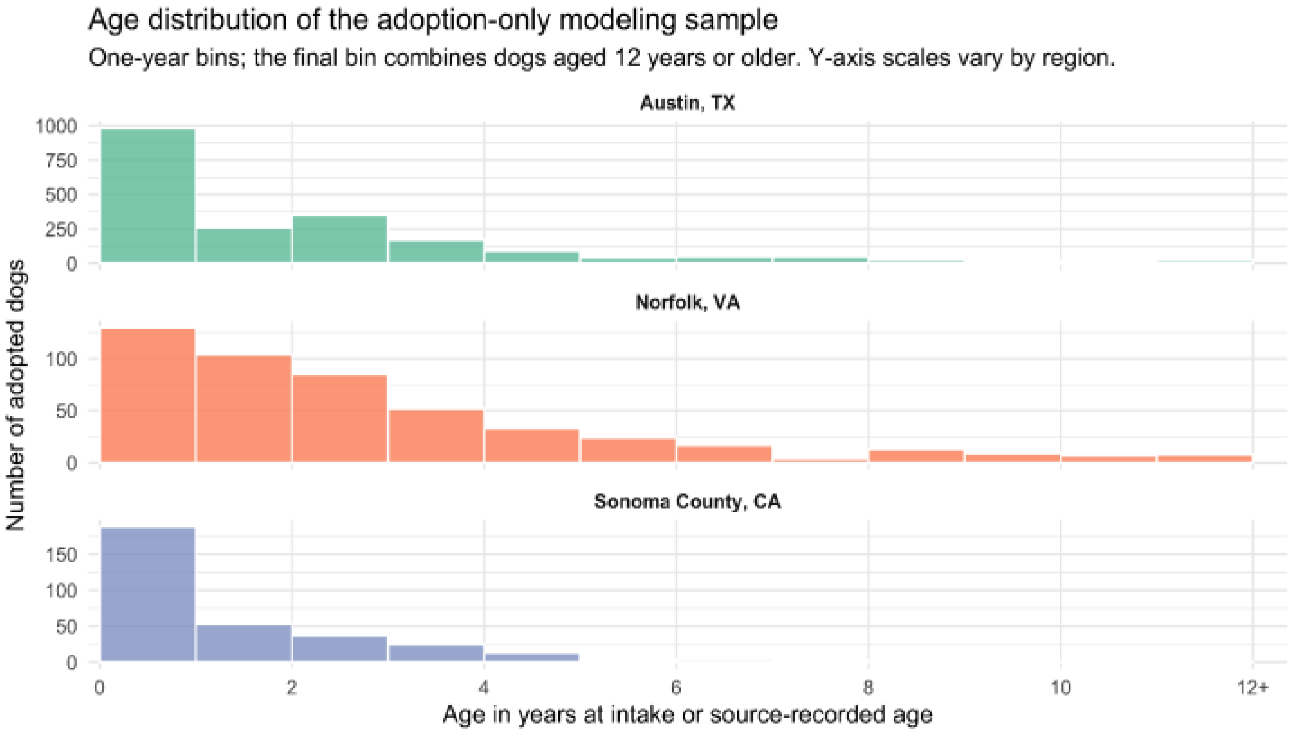
Adopted dogs only; y-axis zoomed to 180 days, boxes use all records.

- Medians: Austin 29, Norfolk 46.5, Sonoma 41 days



Share of age-known dogs with an adoption outcome by each day.

EDA: age by region



- Young dogs make up most adopted dogs in all three regions.
- The age distributions differ slightly between Austin, Norfolk, and Sonoma.

One-year bins; last bin pools ages 12+. Norfolk reports whole years.

Linear regression of log(stay days+1) result

Term	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	3.241	0.237	13.67	<2e-16	
age_groupPuppy	-0.765	0.060	-12.75	<2e-16	***
age_groupYoung	-0.110	0.065	-1.68	0.093	.
age_groupSenior	-0.038	0.109	-0.35	0.728	.
regionNorfolk	-0.224	0.105	-2.13	0.033	*
regionSonoma	0.072	0.129	0.56	0.574	.
sex_groupMale	0.026	0.041	0.64	0.524	.
fixed_statusSpayed	0.611	0.211	2.90	0.004	**
breedSmallCompanion	-1.038	0.087	-12.00	<2e-16	***
breedPit/Staffy	0.234	0.073	3.21	0.001	**
colorWhite	0.032	0.060	0.54	0.590	.
Norfolk:agePuppy	0.827	0.190	4.34	1.45e-05	**

Model fit: N = 2,884 | p < 2.2e-16

- Reference group: Austin adult female, not spayed/neutered, Retriever, Black, January intake
- Positive estimates mean longer predicted log stay; negative estimates mean shorter stay
- Full model includes intake month and all breed/color levels; selected terms shown

ANOVA Test

Ranked term	Df	Sum Sq	F	p value	eta^2	Result
1. Breed family	6	329.76	45.90	6.68e-54	0.088	***
2. Age group	3	207.06	57.64	3.62e-36	0.057	***
3. Region x age	6	36.21	5.04	3.77e-05	0.011	**
4. Region	2	32.53	13.58	1.34e-06	0.010	**
5. Intake Month	11	37.13	2.82	0.0011	0.011	*
6. Color family	7	10.33	1.23	0.281	0.003	.
7. Spay/neuter	1	10.08	8.42	0.0037	0.003	*
8. Sex	1	0.49	0.41	0.524	0.000	.
Residuals	2846	3407.96				

- ANOVA tests each term after accounting for the other model terms
- Top factors: breed family, age group, then intake month
- Sex and color family are not detectable after controls
- Residual variation remains large, so exact-day forecasts stay weak

Ranked category effect

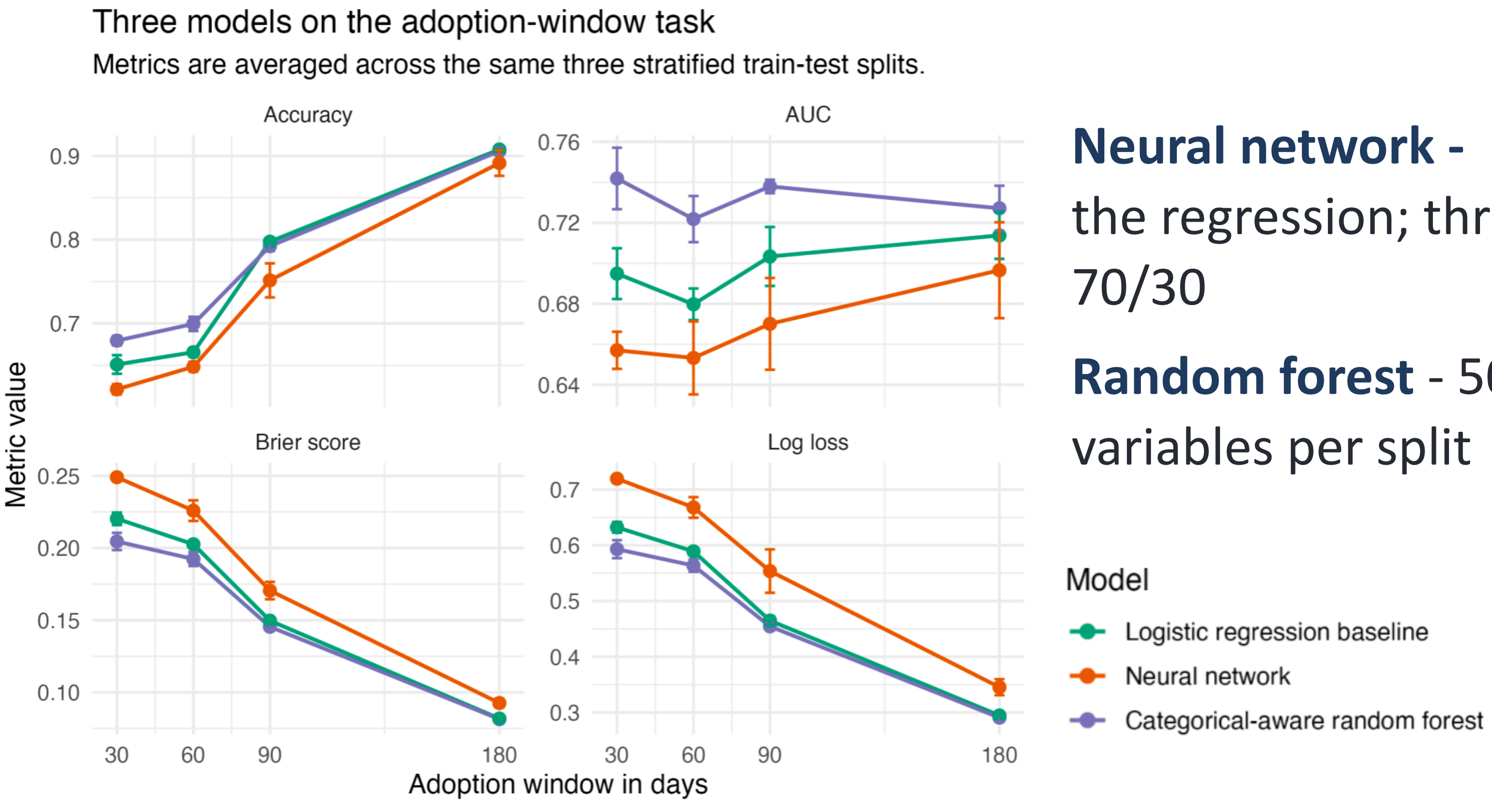
Breed family (reference: Retriever, n = 472)

Rank/category	n	Coef.	% diff	FDR p	Direction
1. Small Companion	293	-1.038	-64.6%	1.3e-31	Shorter
2. Other (n<50)	797	-0.264	-23.2%	2.6e-04	Shorter
3. Pit Bull / Staffy	640	+0.234	+26.3%	0.003	Longer
4. Working / Large	217	-0.084	-8.1%	0.543	
5. Hound / Beagle	102	+0.048	+4.9%	0.833	
6. Shepherd	363	+0.013	+1.3%	0.868	

- Rows are ranked by absolute adjusted coefficient within each predictor block
- % diff = exp(coefficient) - 1, on the log(1 + days) scale
- With Retriever as reference, small companions are ~65% shorter; pit bull/staffy types are ~26% longer

Reference groups: Retriever for breed and Black for color. Same multivariable regression.

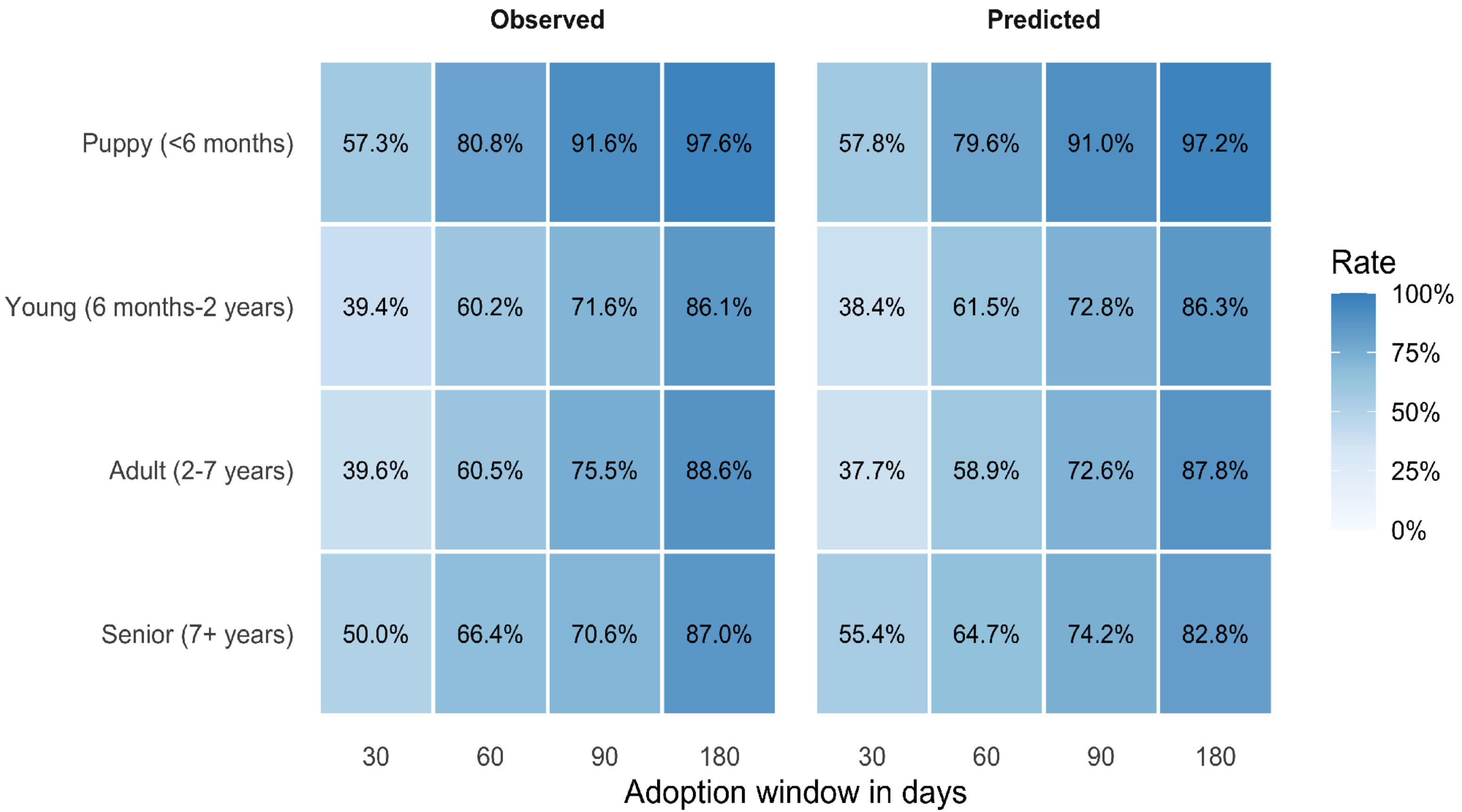
Model Comparison



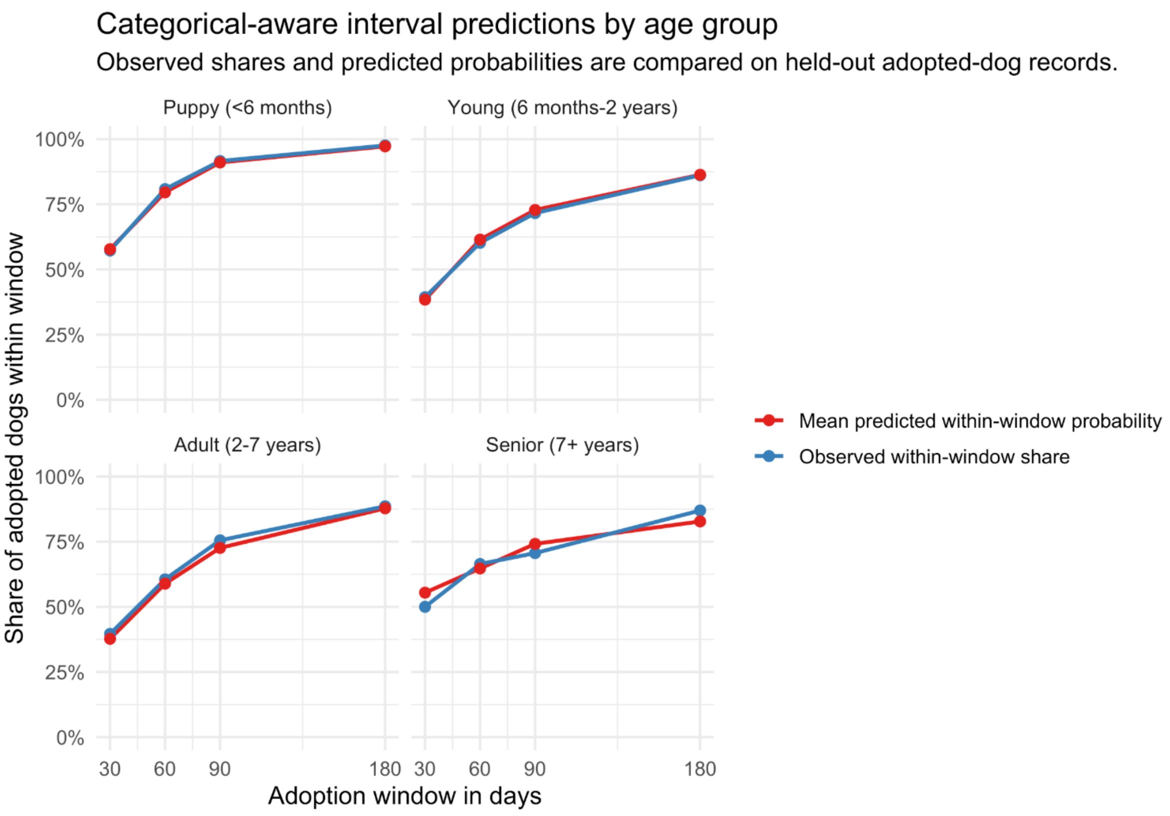
Neural network - Same inputs as the regression; three stratified 70/30

Random forest - 500 trees, 3 variables per split

Random forest: interval predictions

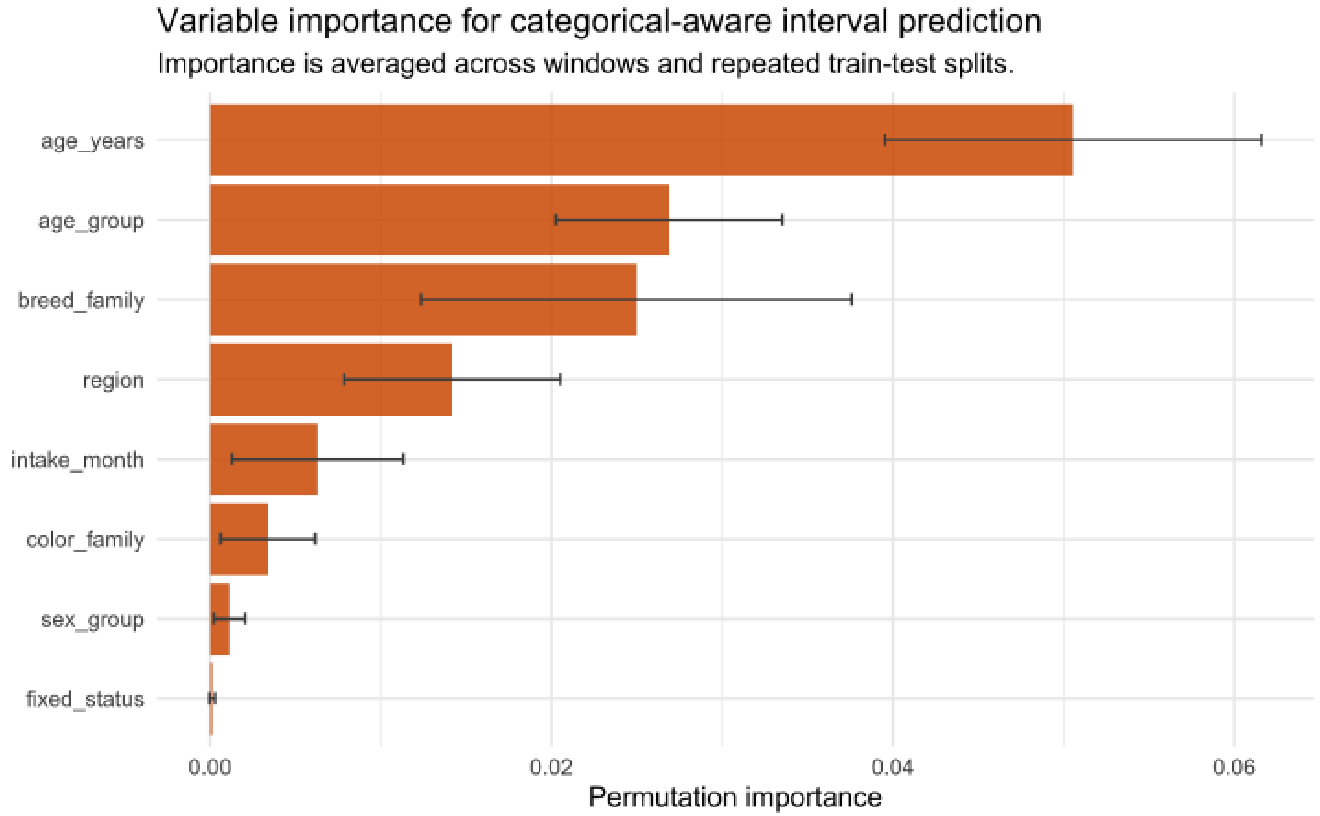


Random forest: interval predictions



Observed and predicted within-window adoption timing rates by age group

Forest calibration and variable importance



Conclusions

What the evidence supports

- Age and breed family have the strongest statistical relationship with speed of adoption
- Exact-day forecasts fail; interval probabilities prove to be more accurate
- Practical use of interval probabilities at intake
  - forecast upcoming kennel crowding
  - strategically plan which dogs will need foster homes

Limitations

- Observational admin records: associations, not causal effects
- Limitations of data
  - missing variable details that halt us from deeper exploration
    - transfers, euthanasia reason
      - dogs killed or transfered are excluded from our study, because of the lack of reasoning
  - lack of health, behavior information
  - lack of foster, capacity, or marketing fields

Future Research

- Impact of health & behavior
- Examination of different shelter policies on marketing and euthanasia
- Impact of adoption fees

AI Usage

- Used AI to find datasets
- Used AI to do basic EDA and data cleaning - distributions, medians by region, and the boxplots
- By providing the overall structure
- Suggested log(1 + days) for the skewed outcome
- Had it read each portal's column documentation and note the difference

Thank You!

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Data sources

• City of Austin Open Data Portal. Austin Animal Center Outcomes. https://data.austintexas.gov/Health-and-Community-Services/Austin-Animal-Center-Outcomes/gsvs-ypi7 (file dated June 30, 2026)

• City of Norfolk Open Data Portal. Norfolk Animal Care and Adoption Center (NACC). https://data.norfolk.gov/Government/Norfolk-Animal-Care-and-Adoption-Center-NACC-/vfm4-5wv6 (file dated July 1, 2026)

• County of Sonoma Open Data Portal. Animal Shelter Intake and Outcome. https://data.sonomacounty.ca.gov/Government/Animal-Shelter-Intake-and-Outcome/924a-vesw (file dated July 9, 2026)

Appendix A: sample dataset rows

Processed sample: adopted dogs only, valid dates, known sex and spay/neuter status, harmonized age, breed, color, and region fields.

Region	Intake	Outcome	Stay	Age	Age group	Breed / color
Austin, TX	2025-04-25	2025-05-06	11.5	3.0	Adult	Other / Black
Austin, TX	2025-04-10	2025-05-06	26.5	3.0	Adult	Other / Patterned
Norfolk, VA	2026-03-22	2026-03-31	9.0	13.0	Senior	Small companion / Tan
Norfolk, VA	2026-03-17	2026-03-26	9.0	3.0	Adult	Small companion / White
Sonoma County, CA	2025-10-12	2025-11-21	40.0	0.7	Young	Shepherd / Black
Sonoma County, CA	2025-09-18	2025-10-28	40.0	0.1	Puppy	Other / White

Final modeling sample: 2,884 adopted dogs = 2,071 Austin, 486 Norfolk, and 327 Sonoma records. Stay length is recomputed as outcome date minus intake date.

Appendix B: testing the region x age interaction

Test logic: compare the structured no-interaction regression with a model adding one interaction block. H0 = the age-group pattern is the same across regions. A small p value supports keeping the interaction.

Candidate	df	dAdjR2	dAIC	F	p	BH p
Region x age group	6	+0.00130	-26.9	5.039	3.77e-05	1.51e-04
Region x breed family	12	+0.0084	-17.0	3.393	5.95e-05	1.59e-04
Age group x breed family	18	+0.0071	-18.6	3.472	9.69e-04	7.75e-03
Region x intake month	22	+0.0009	+18.7	1.132	0.3026	0.3458
Region x color family	14	-0.0008	+16.4	0.817	0.6509	0.6509

Interpretation: the region x age group block is statistically supported and directly matches the research question. Other interactions are treated as supplementary because they answer secondary questions or add many more parameters.